

Plug-and-Play ML Pipelines for SONiC

On-Device Intelligence with Advanced Feature Extraction

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Agenda

1. **Use Cases**
   Where on-device ML applies: app and flow intelligence, device health
2. **SONiC Baseline and the Gap**
   What SONiC telemetry offers today, and what on-device ML still lacks
3. **Feature Extraction from Raw Telemetry**
   Turning raw signals into ML-ready features
4. **Pipeline Architecture**
   Overview and three stages: telemetry ingestion, feature extraction, feature publish
5. **Plug-and-Play and Data Plane Integrity**
   One runtime across sources and engines, running safely on the switch
6. **Implementation Roadmap**
   HFT enhancement, FE HLD, reference implementation, SAI support

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Use Cases

App / Flow Intelligence

- App fingerprinting
- Encrypted Traffic Classification
- Security & Threat detection
- App Experience scoring

Device Health Intelligence

- Link & Device health prediction
- Congestion & Incast anomaly
- Predictive Maintenance
- Intelligent event explanation

Deployment domains **Enterprise** · **Datacenter** · **Management & Control**

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Baseline: SONiC today

The raw signals are already there. The path from telemetry to on-device ML is not.

What SONiC has today

- Telemetry plumbing: COUNTERS_DB, IPFIX, sFlow, gNMI
- Rich port, queue and PHY counters
- Packet sampling (psample) and mirroring
- SAI abstraction across ASICs

What's missing for on-device ML

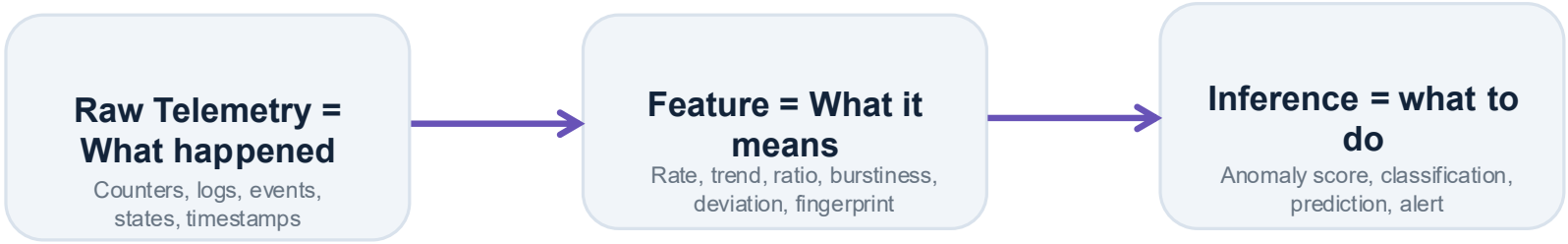
- Most ML analytics need feature vectors; raw telemetry isn’t sufficient
- Common interface to feed models, on-switch or off-switch
- Flexible framework allowing optional actions by models

This design adds the missing path: Extract features on the device, Feed them to models through one common interface.

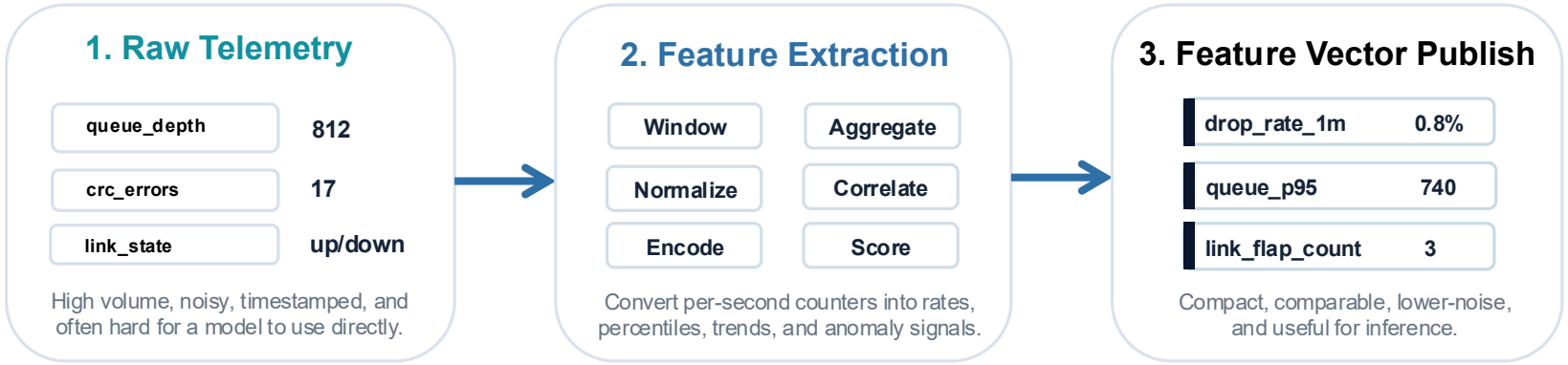
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Feature Extraction from Raw Telemetry

Feature Extraction = Extraction of ML-Ready Signals

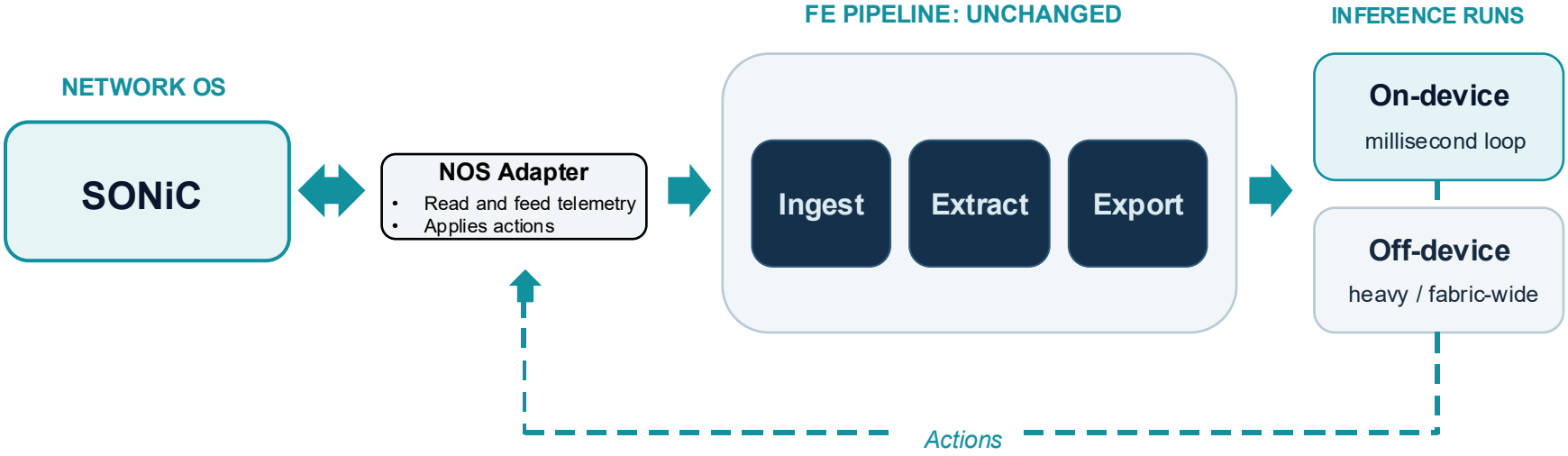


Feature Extraction Pipeline



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Feature Extraction Architecture



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Stage 1 · Telemetry Ingestion

Heterogeneous signals to application defined features

Redis COUNTERS_DB	**Polled at interval**	Port, queue, and PFC counters read from the SONiC state store.
IPFIX	**Time-windowed flow aggregates**	Per-flow counts/bytes and timestamps summed over a time window.
Packet capture	**Sampled or mirrored**	Sampled (sFlow, psample) or full copies (ACL/ERSPAN mirror, first-N of flow).

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Stage 2 · Feature Extraction

Windowing

Bounded time / count windows over the event stream.

Normalization

Scale and standardize so features are model-ready.

Transforms

Derive rates, ratios, and aggregates into a vector.

feature vector

Types of Features

Device Health Intelligence
SNR, FEC, optics PHY & link errors.

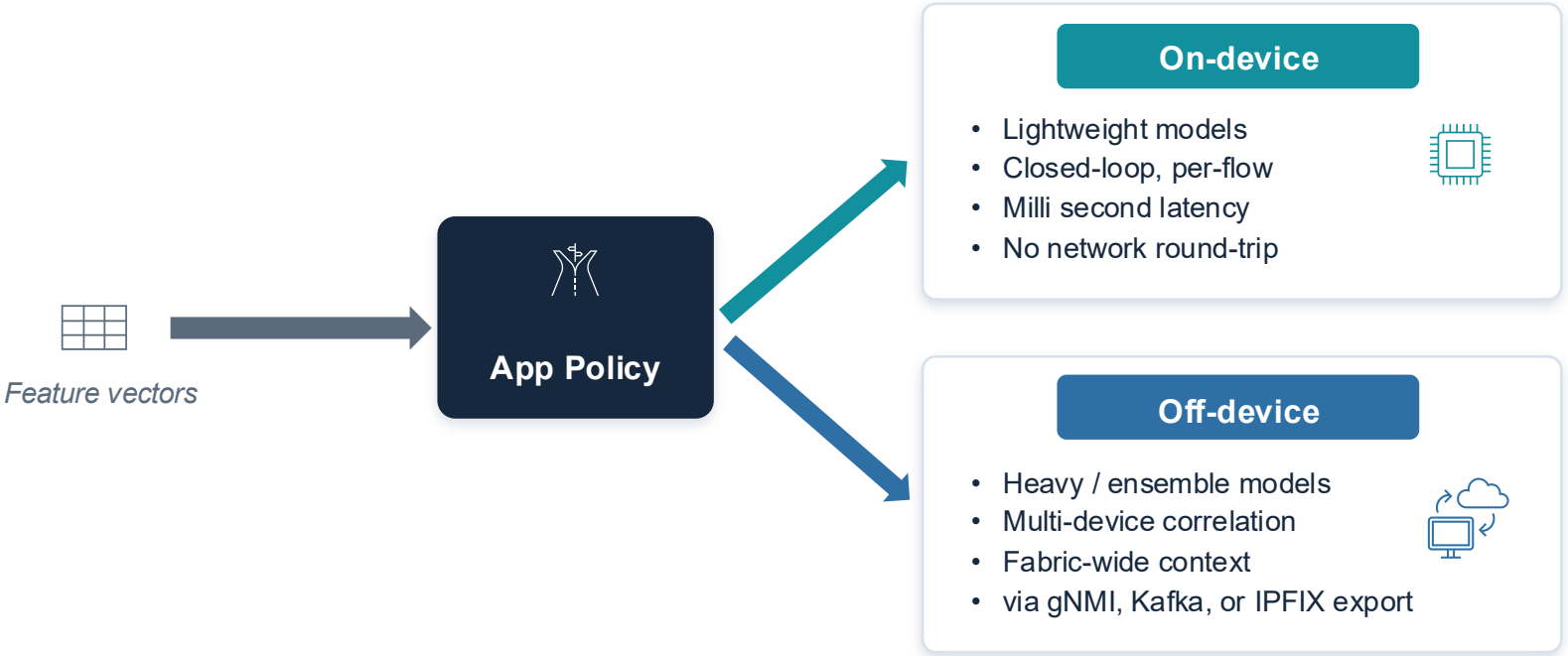
App / Flow Intelligence
Per-flow records, IPFIX / sFlow.

App chooses the required features: the framework assumes nothing about the use-case

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Stage 3 · Feature Publish

Routes vectors to inference as per application policy and latency needs



Same vectors, different Destination: chosen per application, with no change to pipeline code.

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One pipeline for Every Model

The pipeline is agnostic to the features you select.

Any source works



COUNTERS_DB, IPFIX, and sFlow all land on one event bus - extraction never sees the source.

Any engine works



ONNX Runtime, TFLite, or your own: same interface.

Runs anywhere



On-switch or off-switch is a config choice: the pipeline code never changes.

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Preserving Data Plane Integrity

Forwarding-path integrity

- Runs in an isolated container with cgroup CPU and memory caps
- Uses standard NOS interfaces
- Doesn’t starve control-plane processes like BGP under load

NOS lifecycle

- Survives warm reboot, fast reboot, and config reload
- In-memory state is size-capped and saved, so it can’t grow without limit
- Starts, stops, and upgrades cleanly like any managed service

It safely runs alongside Dataplane Plane functions with ZERO effect

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Implementation Roadmap



Enhance SONiC's HFT to support flow telemetry

Proposal for Feature Extraction(FE) HLD in SONiC

Reference Implementation of Feature Extraction for SONiC Virtual Switch

Release SAI support for Feature Extraction on Marvell® platforms

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Questions & Discussion

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